

Code-Layout, Readability and Reusability

Date	25 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	3 Marks

Code-Layout, Readability and Reusability:

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing used throughout the code	Yes	PEP-8 followed
2	Proper File Structure	Files and folders logically organized	Yes	MVC structure
3	Meaningful Variable Names	Variables clearly indicate purpose	Yes	Readable names
4	Function / Method Names	Functions descriptively named	Yes	Easy maintenance
5	Code Comments	Important logic documented	Yes	Inline comments
6	Modular Design	Reusable modules for preprocessing, training and prediction	Yes	Reusable architecture
7	No Redundant Code	Duplicate code removed	Yes	Optimized
8	Error Handling	Input validation and exception handling	Yes	User-friendly errors

Reusable Components / Modules:

S.No	Component / Module	Language / Technology	Where Reused	Reusability
1	Data Preprocessing	Python, Pandas	Training & Prediction	High
2	ML Prediction	Scikit-learn	Prediction Engine	High
3	Visualization	Matplotlib, Seaborn	Dashboard	High
4	Flask Routes	Flask	Web Application	High
5	Input Validation	Python	Forms	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized project
Readability	5	Clear naming and comments
Reusability	5	Reusable modules
Documentation / Comments	5	Adequate documentation
Overall Score	5/5	Maintainable and scalable